

Midterm Assessment: Student Exam Performance

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Part 1: Audience Focused Executive Summary

Purpose

I was asked to analyze exam results for 1,000 students and explain what separates higher performers from lower ones. The aim is to give decision makers a clear picture of where extra support would help most. I looked at three subjects alongside background details such as gender, parental education, whether a student completed a test preparation course, and which lunch program they were on.

What I found

Three patterns stand out. First, preparation pays off. Students who completed the test preparation course scored higher in every subject, about 7.6 points higher on average. This is encouraging, because it is something a school can directly influence. Second, economic background appears to matter even more. Students on the standard lunch program scored about 8.6 points higher than students on the free or reduced lunch program, the largest gap I found. Lunch status is commonly used as an indicator of household income, so this gap likely reflects financial circumstances at home rather than anything about the students ability. The gap was widest in math. Third, reading and writing go hand in hand. A student who reads well almost always writes well; the two scores rise and fall together almost perfectly. Math is related too, but a little more independent. Gender differences were smaller and depended on the subject: girls scored higher in reading and writing, while boys scored slightly higher in math.

Recommendations

- Expand access to the test preparation course, since it helped across every subject and is within the school's control.
- Direct additional academic support toward students on the free or reduced lunch program, where the largest gap appears, recognizing that some of this support may need to address needs outside the classroom.
- Treat reading and writing as a single literacy focus, because improving one tends to lift the other, which makes support more efficient.

Part 2: Exploratory Data Analysis

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style="whitegrid")
```

```
df = pd.read_csv("StudentsPerformance.csv")
df.head()
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score	
0	female	group B	bachelor's degree	standard	none	72	72	74	
1	female	group C	some college	standard	completed	69	90	88	
2	female	group B	master's degree	standard	none	90	95	93	
3	male	group A	associate's degree	free/reduced	none	47	57	44	
4	male	group C	some college	standard	none	76	78	75	

Next steps: [Generate code with df](#) [New interactive sheet](#)

The columns

- gender - female or male

- race/ethnicity - anonymized group label (A to E)
- parental level of education - highest education of a parent
- lunch - standard, or free/reduced (a common stand-in for household income)
- test preparation course - completed or none (our stand-in for study habits)
- math / reading / writing score - exam results from 0 to 100

```
df.info()
print()
print(df.isnull().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   gender                                1000 non-null   object
 1   race/ethnicity                        1000 non-null   object
 2   parental level of education           1000 non-null   object
 3   lunch                                 1000 non-null   object
 4   test preparation course               1000 non-null   object
 5   math score                            1000 non-null   int64
 6   reading score                         1000 non-null   int64
 7   writing score                         1000 non-null   int64
dtypes: int64(3), object(5)
memory usage: 62.6+ KB
```

```
gender                0
race/ethnicity         0
parental level of education  0
lunch                  0
test preparation course  0
math score              0
reading score           0
writing score           0
dtype: int64
```

The dataset has no missing values, next we rename the columns to remove spaces and slashes, and we create a single `average_score` column to summarize each student with one number.

```
df = df.rename(columns={
    "race/ethnicity": "race_ethnicity",
    "parental level of education": "parental_education",
    "test preparation course": "test_prep",
    "math score": "math",
    "reading score": "reading",
    "writing score": "writing",
})

df["average_score"] = df[["math", "reading", "writing"]].mean(axis=1)
df.head()
```

	gender	race_ethnicity	parental_education	lunch	test_prep	math	reading	writing	average_score	
0	female	group B	bachelor's degree	standard	none	72	72	74	72.666667	
1	female	group C	some college	standard	completed	69	90	88	82.333333	
2	female	group B	master's degree	standard	none	90	95	93	92.666667	
3	male	group A	associate's degree	free/reduced	none	47	57	44	49.333333	
4	male	group C	some college	standard	none	76	78	75	76.333333	

Next steps: [Generate code with df](#) [New interactive sheet](#)

Summary statistics

```
df.describe()
```

	math	reading	writing	average_score
count	1000.00000	1000.000000	1000.000000	1000.000000
mean	66.08900	69.169000	68.054000	67.770667
std	15.16308	14.600192	15.195657	14.257326
min	0.00000	17.000000	10.000000	9.000000
25%	57.00000	59.000000	57.750000	58.333333
50%	66.00000	70.000000	69.000000	68.333333
75%	77.00000	79.000000	79.000000	77.666667
max	100.00000	100.000000	100.000000	100.000000



```
for col in ["math", "reading", "writing", "average_score"]:
    print(col)
    print("  mean    ", round(df[col].mean(), 2))
    print("  median  ", round(df[col].median(), 2))
    print("  variance", round(df[col].var(), 2))
```

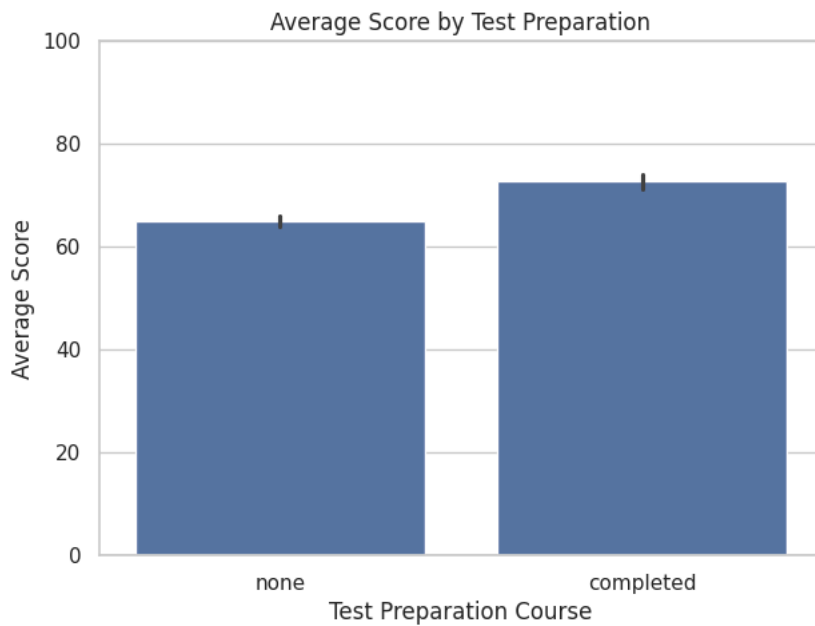
```
math
mean    66.09
median  66.0
variance 229.92
reading
mean    69.17
median  70.0
variance 213.17
writing
mean    68.05
median  69.0
variance 230.91
average_score
mean    67.77
median  68.33
variance 203.27
```

Average scores sit in the high 60s. Medians are almost identical to the means, so the scores are fairly symmetric rather than skewed. Writing and math show the widest spread, while the combined average is the most stable because averaging three subjects smooths out individual swings.

Chart 1: Categorical comparison

Does completing the test preparation course relate to higher scores?

```
plt.figure(figsize=(7, 5))
sns.barplot(data=df, x="test_prep", y="average_score")
plt.title("Average Score by Test Preparation")
plt.xlabel("Test Preparation Course")
plt.ylabel("Average Score")
plt.ylim(0, 100) # honest baseline at zero
plt.show()
```

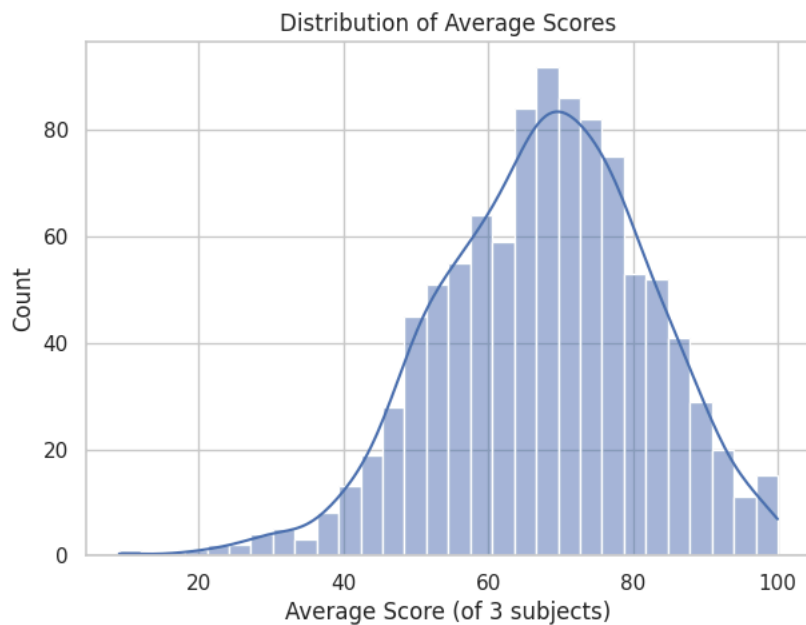


Students who completed the preparation course average about 72.7, versus 65.0 for those who did not, a gap of roughly 7.6 points. The gain holds in every subject and is largest in writing. The y-axis starts at zero so the bar heights represent the scores honestly.

Chart 2: Distribution

How are overall scores spread across all students?

```
plt.figure(figsize=(7, 5))
sns.histplot(data=df, x="average_score", kde=True, bins=30)
plt.title("Distribution of Average Scores")
plt.xlabel("Average Score (of 3 subjects)")
plt.show()
```



The distribution is roughly bell-shaped and centered near 68, with most students between about 55 and 80. A thin left tail shows a small number of low performers, which is where targeted support would matter most.

Chart 3: Relationship

Do reading and writing scores move together?

```
plt.figure(figsize=(7, 5))
sns.regplot(data=df, x="reading", y="writing",
            scatter_kws={"alpha": 0.4}, line_kws={"color": "red"})
plt.title("Reading vs Writing Scores")
plt.xlabel("Reading Score")
plt.ylabel("Writing Score")
plt.show()
```



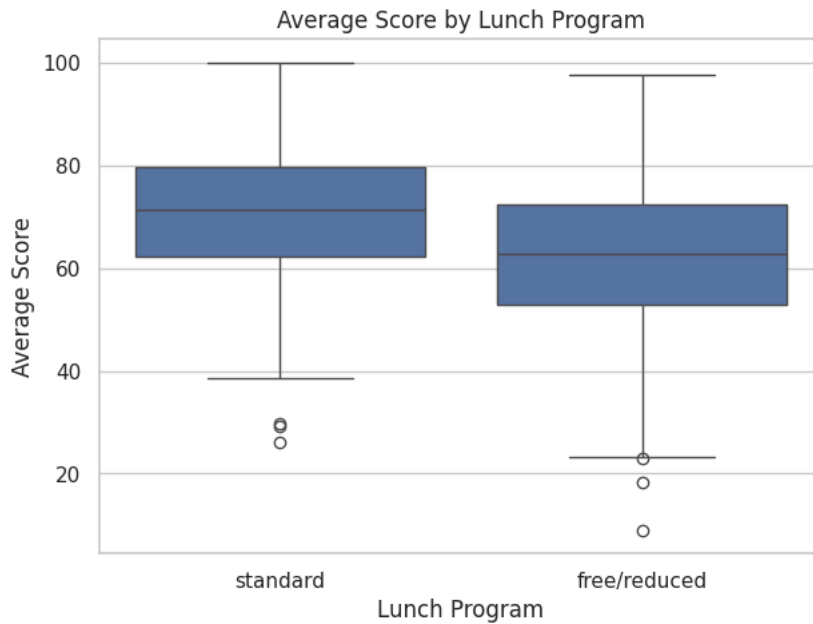
```
df[["math", "reading", "writing"]].corr().round(3)
```

	math	reading	writing
math	1.000	0.818	0.803
reading	0.818	1.000	0.955
writing	0.803	0.955	1.000

Reading and writing are very strongly correlated, so a student strong in one is almost always strong in the other. Math is also strongly related to both but a little more independent. In practice, reading and writing partly measure the same underlying literacy skill.

Chart 4: Score by lunch program

```
plt.figure(figsize=(7, 5))
sns.boxplot(data=df, x="lunch", y="average_score")
plt.title("Average Score by Lunch Program")
plt.xlabel("Lunch Program")
plt.ylabel("Average Score")
plt.show()
```



Students on the standard lunch average about 70.8, versus 62.2 for those on free or reduced lunch, a gap of roughly 8.6 points, the largest of any factor here. The gap is widest in math (about 11 points). Because lunch status reflects economic background, this points to support being needed beyond the classroom.

Part 3: Visualization Critique

```
# Fake data
categories = ['A', 'B', 'C', 'D']
values = [95, 96, 97, 98]

# Terrible chart
plt.figure(figsize=(10, 2))

plt.bar(categories, values,
        color=['red', 'lime', 'cyan', 'magenta'])

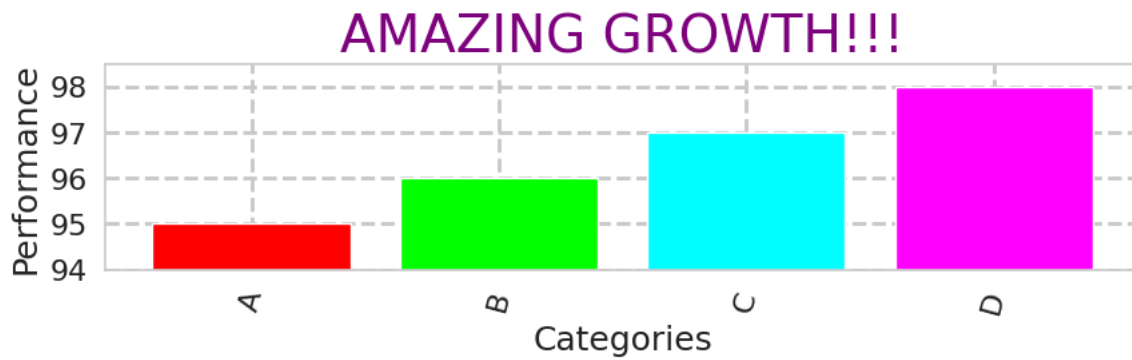
# Misleading axis
plt.ylim(94, 98.5)

# Bad styling choices
plt.title("AMAZING GROWTH!!!", fontsize=28, color='purple')
plt.xlabel("Categories", fontsize=18)
plt.ylabel("Performance", fontsize=18)

# Hard-to-read grid
plt.grid(True, linestyle='--', linewidth=2)

# Rotate labels unnecessarily
plt.xticks(rotation=75, fontsize=16)
plt.yticks(fontsize=16)

plt.show()
```



What makes this visualization misleading

The biggest problem is the y-axis. It starts at 94 instead of 0, so values that differ by only about one percent (95 to 98) appear to differ enormously; the last bar looks several times taller than the first when the four values are nearly identical. The title, AMAZING GROWTH, makes things worse by announcing a conclusion the data does not support and adding emphasis that has no place on a chart. The four neon colors carry no meaning, since the categories are not ranked or grouped.

Which visualization principles are violated

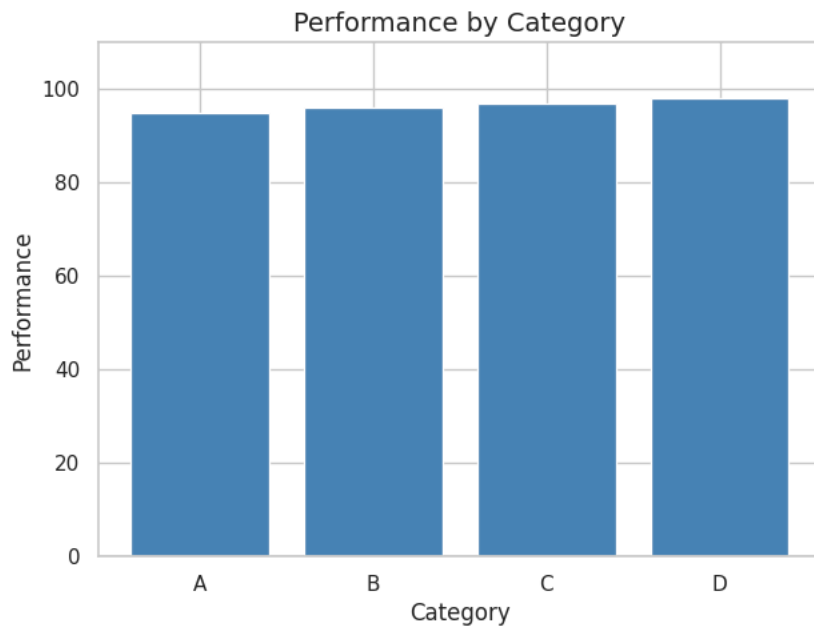
- A bar's length should be proportional to its value, which truncating the axis breaks directly.
- The thick dashed gridlines and oversized fonts are clutter that competes with the data.
- The chart editorializes through its title instead of letting the reader judge.
- Rotating the labels 75 degrees and using arbitrary colors makes the chart harder to read for no analytical reason.

How I would redesign it

I would start the y-axis at 0 so the bars tell the truth about magnitude, use a single muted color since the categories need no color coding, replace the title with a neutral description such as Performance by Category, set the labels horizontal, and remove the heavy gridlines.

```
categories = ['A', 'B', 'C', 'D']
values = [95, 96, 97, 98]

plt.figure(figsize=(7, 5))
plt.bar(categories, values, color="steelblue")
plt.ylim(0, 110)  # honest baseline at zero
plt.title("Performance by Category", fontsize=14)
plt.xlabel("Category")
plt.ylabel("Performance")
plt.show()
```



Part 4: Technical Report

Introduction

Schools want to understand which factors are associated with stronger exam performance so they can target limited support effectively. This report analyzes a dataset of 1,000 students, each with scores in three subjects (math, reading, and writing) and several background attributes: gender, race/ethnicity (anonymized), parental level of education, lunch program status, and whether the student completed a test preparation course. The question I set out to answer is which of these factors relate most clearly to performance, and what a school might do in response.

Methods

Data preparation. I loaded the dataset and confirmed it contained no missing values, so no imputation was needed. I renamed the columns to remove spaces and slashes for cleaner code, and I created an `average_score` column equal to the mean of the three subject scores, giving one summary measure per student.

Analysis approach. I computed summary statistics (mean, median, and variance) for each subject and for the combined average. I then examined the data through four chart types: a categorical comparison (average score by test preparation), a distribution (the spread of average scores), a relationship (reading versus writing), and a correlation view, plus an additional chart for gender. All work was done in Python using pandas, seaborn, and matplotlib.

Results

Summary statistics show average scores in the high 60s (math 66.1, reading 69.2, writing 68.1) with medians almost equal to the means, showing roughly symmetric distributions. The categorical comparison (Chart 1) shows students who completed the test preparation course averaging 72.7 against 65.0 for those who did not, a difference of about 7.6 points that holds across all three subjects. The distribution (Chart 2) is approximately bell-shaped and centered near 68, with a thin left tail of low performers. The relationship chart (Chart 3) and the correlation table show reading and writing are very strongly correlated ($r = 0.96$), while math correlates with each at about 0.80. The lunch comparison (Chart 4) reveals the largest gap in the data: standard-lunch students average 70.8 versus 62.2 for free or reduced lunch, a difference of about 8.6 points, widest in math.

Discussion

Two factors stand out as most associated with performance: lunch program status and test preparation. Because lunch status is a standard proxy for household income, its large effect suggests that economic circumstances outside school weigh heavily on results, particularly in math. Test preparation, by contrast, is something the school controls directly, and its consistent benefit across subjects makes it an appealing lever. The very high correlation between reading and writing implies the two are partly measuring the same underlying literacy skill, so they can be supported together. Throughout, it is important to stress that these are observational associations.

The data cannot establish cause, and unmeasured factors (school quality, home environment, prior achievement) may drive several of these patterns at once.